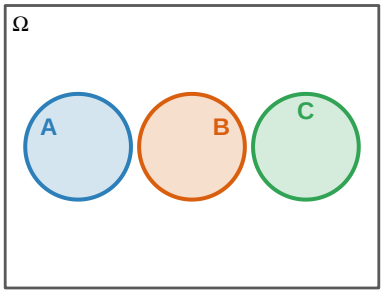
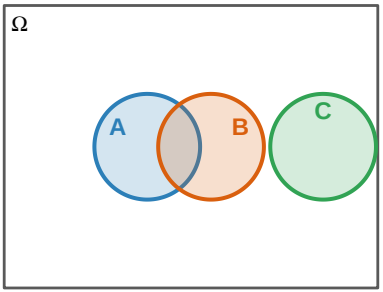
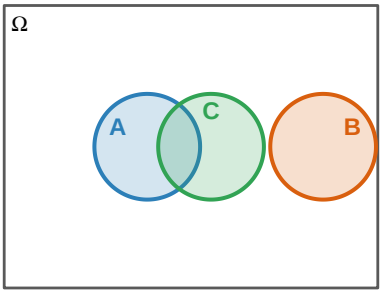
1. All three pairs are disjoint



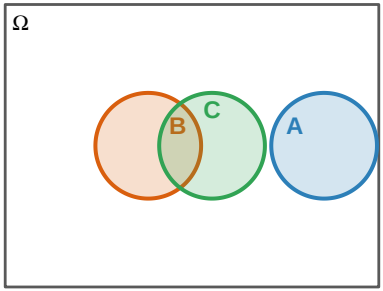
2. Only A and B overlap



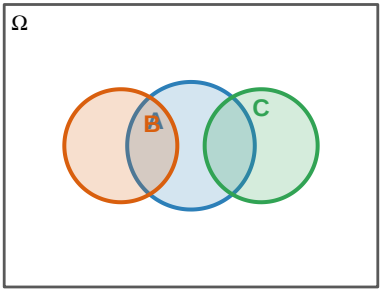
3. Only A and C overlap



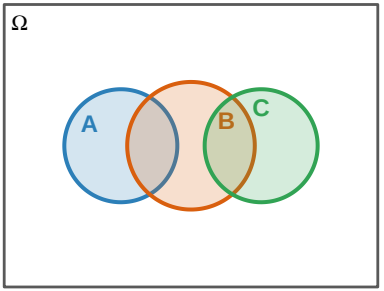
4. Only B and C overlap



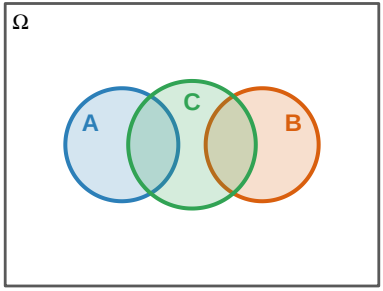
5. AB and AC overlap



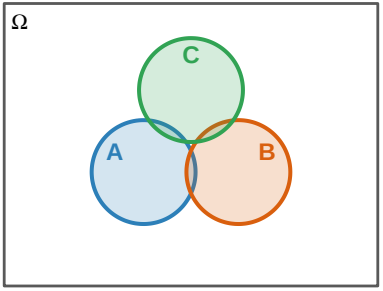
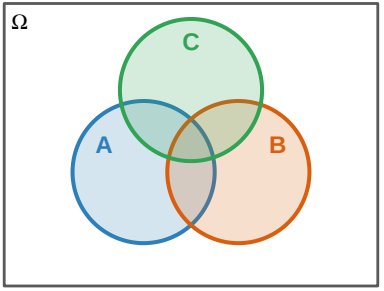
6. AB and BC overlap



7. AC and BC overlap



8. Every pair overlaps; no triple overlap



9. A common triple overlap